SIMATS ENGINEERING

C03 ASSESSMENT TOOL1

COURSE CODE: CSA1610 **COURSE NAME: DWDM**

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EXPERIMENTS

Q1 R Code for Covariance and Correlation:

CODE:

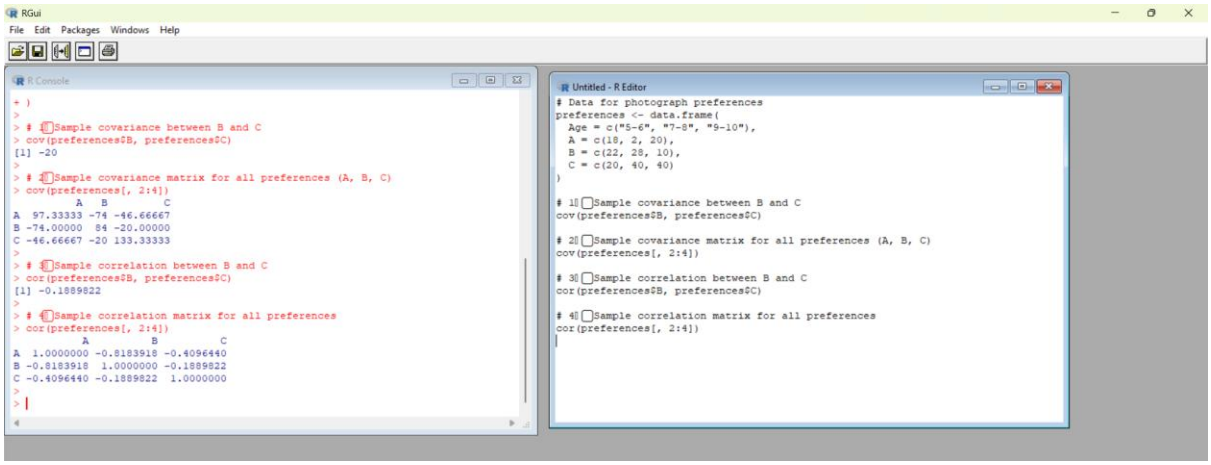
```
# Data for photograph preferences
preferences <- data.frame(
  Age = c("5-6", "7-8", "9-10"),
  A = c(18, 2, 20),
  B = c(22, 28, 10),
  C = c(20, 40, 40)
)

# Sample covariance between B and C
cov(preferences$B, preferences$C)

#  Sample covariance matrix for all preferences (A, B, C)
cov(preferences[, 2:4])

#  Sample correlation between B and C
cor(preferences$B, preferences$C)

# Sample correlation matrix for all preferences
cor(preferences[, 2:4])
```



Q2 R Code for Boxplot and Outlier Detection

CODE:

```
# Data: Points scored by players

points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)


# Create a boxplot

boxplot(points,

    main = "Tennis Players Points Distribution",

    ylab = "Points Scored",

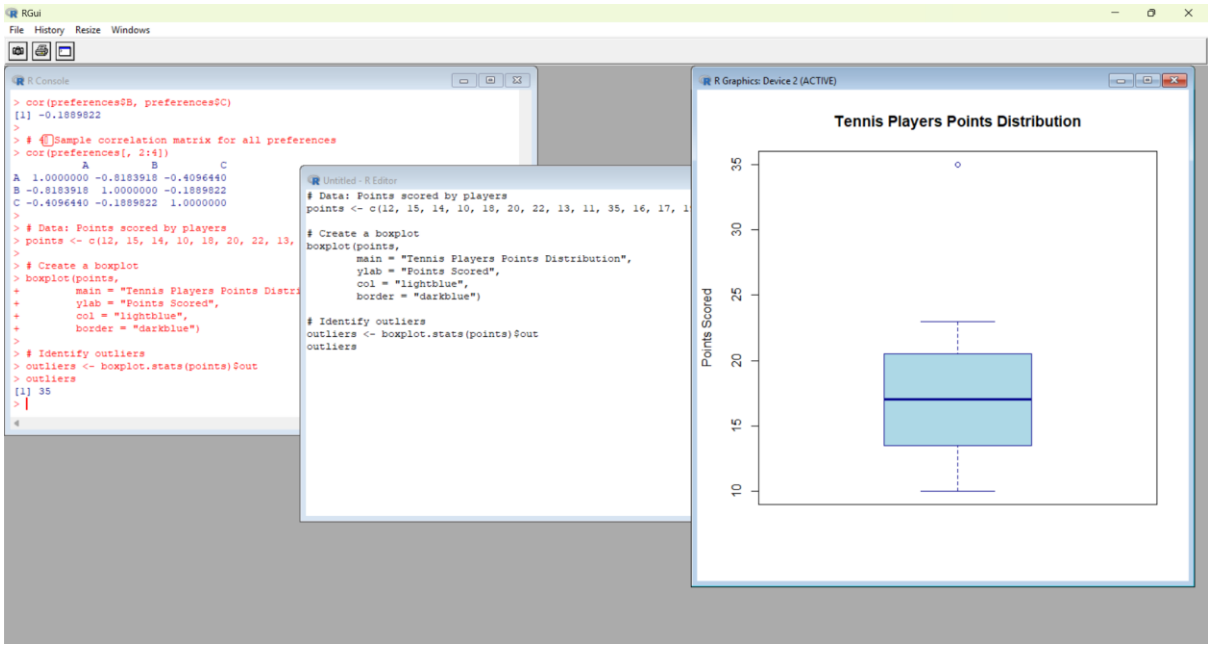
    col = "lightblue",

    border = "darkblue")


# Identify outliers

outliers <- boxplot.stats(points)$out

outliers
```



Q3 R Code for Decision Tree Classifier

CODE:

```
# Install and load required packages

# install.packages("rpart")

# install.packages("rpart.plot")

library(rpart)

library(rpart.plot)


# Dataset

loan_data <- data.frame(

 Age = c("young","young","middle","old","old","old","middle","young","young","old",

       "middle","middle","old","young","middle"),

 Income = c("high","high","high","medium","low","low","low","medium","low","medium",

        "medium","high","medium","medium","low"),

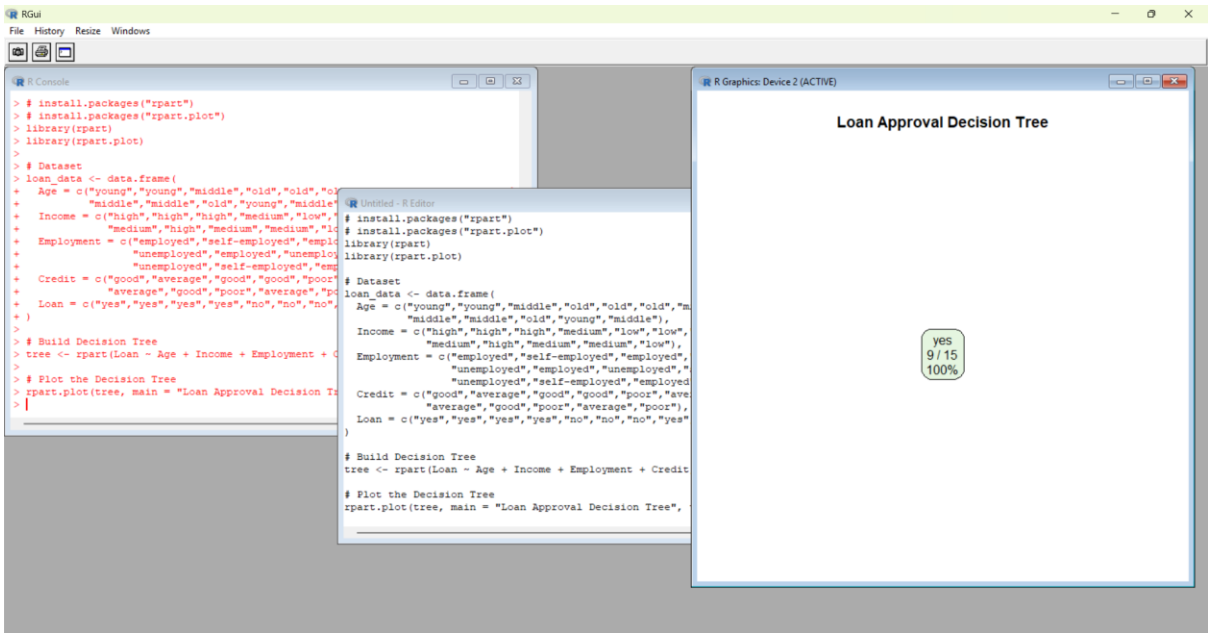
 Employment = c("employed","self-
employed","employed","employed","unemployed","self-employed",

            "unemployed","employed","unemployed","self-employed","employed","self-
employed",
```

```
            "unemployed","self-employed","employed"),
  Credit = c("good","average","good","good","poor","average","poor","average","poor","good",
             "average","good","poor","average","poor"),
  Loan = c("yes","yes","yes","yes","no","no","no","yes","no","yes","yes","yes","no","yes","no")
)

# Build Decision Tree
tree <- rpart(Loan ~ Age + Income + Employment + Credit, data = loan_data, method =
"class")

# Plot the Decision Tree
rpart.plot(tree, main = "Loan Approval Decision Tree", type = 3, extra = 102)
```



Q4 R Code for Mean, Median, Range, and Boxplot

CODE:

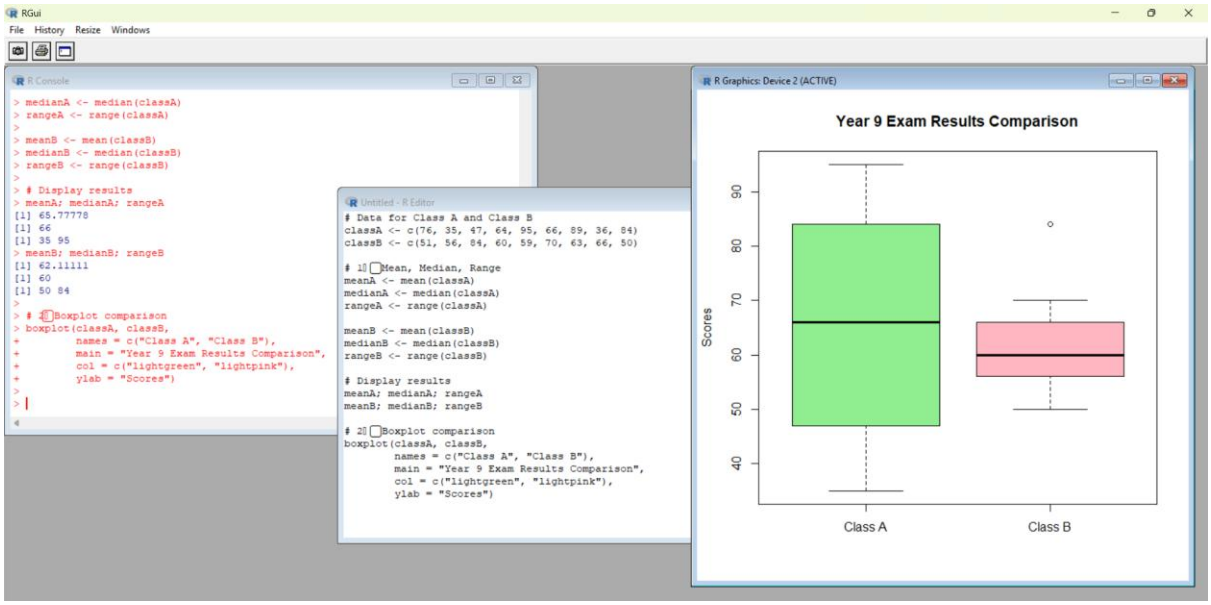
```
# Data for Class A and Class B
classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
```

```
# Mean, Median, Range
meanA <- mean(classA)
medianA <- median(classA)
rangeA <- range(classA)

meanB <- mean(classB)
medianB <- median(classB)
rangeB <- range(classB)

# Display results
meanA; medianA; rangeA
meanB; medianB; rangeB

#  Boxplot comparison
boxplot(classA, classB,
      names = c("Class A", "Class B"),
      main = "Year 9 Exam Results Comparison",
      col = c("lightgreen", "lightpink"),
      ylab = "Scores")
```



Q5 R Code for Statistical Analysis

CODE:

```
# Data

age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61)

fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2, 34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)


# (a) Mean, Median, Standard Deviation

mean(age); median(age); sd(age)

mean(fat); median(fat); sd(fat)


# (b) Boxplots

par(mfrow = c(1, 2))  # Display side-by-side

boxplot(age, main = "Boxplot of Age", col = "lightblue", ylab = "Age")

boxplot(fat, main = "Boxplot of Body Fat %", col = "lightgreen", ylab = "% Fat")


# (c) Scatter Plot and Q-Q Plot

par(mfrow = c(1, 2))
```

plot(age, fat, main = "Scatter Plot: Age vs Body Fat", xlab = "Age", ylab = "% Fat", pch = 19, col = "darkblue")

qqplot(age, fat, main = "Q-Q Plot: Age vs Body Fat", xlab = "Age Quantiles", ylab = "Body Fat Quantiles", col = "purple")

